

TABLE III
Description of Packages

Name of the package	Description
bitstreamTools.bitstream	Represents the packet view of a Xilinx bitstream. Contains export facilities and classes to represent header, packets, types, configuration registers and bitstream parsing.
bitstreamTools.bitstream.test	Contains classes and scripts to test the bitstream package.
bitstreamTools.configuration	Provides FPGA-level view of configuration data in a bitstream using frames. Contains an implementation of frame address register.
bitstreamTools.configurationSpecification	Contains specifications, in column layouts, of all supported devices. Defines different constructs such as block types, block sub-types and part library functions.
bitstreamTools.examples	Provides some examples on how to use the bitstream functionality in RapidSmith.
bitstreamTools.examples.support	Gives some support classes for the examples in the previous bitstreamTools.examples package.
design	Represents all of the constructs in XDL design files, for example, Nets, Instances, PIPs, Modules, and Designs.
design.explorer	A GUI interactive explorer that allows the user to navigate through the various constructs in the design such as instances, Nets, Modules and Module Instances. Has a tile map which allows the user to view the locations of various objects on the FPGA fabric. Also contains an experimental timing report parser to correlate timing information with a design.
design.parser	A JavaCC-based parser for XDL files which populates an instance of the Design class in the design package.
device	Encloses all details of an FPGA device, such as part name, tiles, primitive sites and routing resources. All information about Xilinx parts is populated in device from the XDLRC files generated by xdl executable.
device.brower	An extension of the part tile browser of the examples package. Allows the user to browse all of the installed parts and also navigate primitive sites as well as routing resources.
device.helper	Comprises some classes to help in the creation of the device files.
examples	Contains a few examples for users on how to use RapidSmith.
gui	It is used to help build graphical programs in Qt Jambi for RapidSmith. Contains some commonly used widgets that can be put together easily using the Qt Jambi framework.
placer	Encompasses classes to place designs.
router	Contains classes to route design. Has a framework to help the RapidSmith users to create new routers.
tests	Comprises test classes that would exercise various portions of RapidSmith.
timing	Currently, this package is an experimental TWR parser that will parse timing reports output from Xilinx Trace (trce).
util	Miscellaneous support classes and utilities, including the installer.

given design in XDL format. Consider the simple design browser tool given in Fig. 1. It is a screenshot of a tree-based graphical list traversal of an XDL design file. Here you can see primitive instances, pins and routing protective Internet

protocols (PIPs). The screenshot of a similar tool is provided in Fig. 2, which shows a program using RapidSmith, with the ability to interactively search for primitive instances in a loaded design. Yet another example is the design

analyser which tabulates detailed resource utilisation statistics and design properties.

Design creation tool

Apart from reading and writing XDL files, RapidSmith's APIs provide the user with a mechanism to create a circuitry in XDL file format. It also has API routines to create primitive instances,

customise these instances and even create logical connections between primitive instance pins.

Physical design tools

This software offers an excellent framework for the creation of physical design tools, for example, placement and routing tools. Fig. 3 shows the screenshot of an interactive hard macro placement tool built on RapidSmith.

In short, RapidSmith is a software that lets you manipulate XDL file formats, and provides a great platform on which one can develop a complete FPGA design tool suite. It allows for a variety of circuit manipulations. It opens the door to new opportunities for research in FPGA CAD tools. ●

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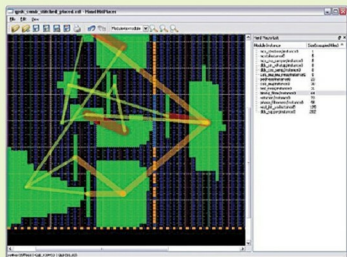


Fig. 3: Screenshot of an interactive hard macro placement tool